

Maka Paka Symmetraca

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

Zena has a line of n boxes, where each box contains an integer. She wants the line to become **symmetric**.

A line is called symmetric if for every index i ($0 \leq i < \lfloor n/2 \rfloor$), the values in boxes i and $n - 1 - i$ are equal. The order of boxes cannot be changed. If n is odd, the middle box is ignored.

Your task is to make the line symmetric with the **minimum total cost**.

Step 1: Symmetrizing the original boxes

For each mirrored pair of boxes (a_i, a_{n-1-i}) , Zena must choose **exactly one** of the following operations:

1. Increase or decrease the value of either box by 1 any number of times until both values become equal. The cost is equal to the number of operations performed.
2. Replace **both** values with their absolute difference $|a_i - a_{n-1-i}|$. The cost of this operation is $2 \times |a_i - a_{n-1-i}|$.

For each mirrored pair, Zena always chooses the operation with the **minimum cost**. The chosen cost for every pair is stored in a new array called **diffs**.

Step 2: Symmetrizing the cost array

Let m be the length of the array **diffs**. Zena now treats **diffs** as a new line of boxes.

For every mirrored pair $(\text{diffs}[i], \text{diffs}[m - 1 - i])$, she may increase or decrease the value of either element by 1 until both values become equal. The cost is equal to the number of operations performed.

Total Cost = (cost from Step 1) + (cost from Step 2).

Input

The first line contains a single integer n ($3 \leq n \leq 1000$), the number of Boxes The second line contains n integers, call each element as x , then ($1 \leq x \leq 10$).

Output

output a single line contains the total cost as described in the statement.

Example

standard input	standard output
5 1 2 3 4 5	8

Note

Input: 5

1 2 3 4 5

Step 1: Symmetrize the original Boxes Mirrored pairs and their absolute differences:

- $(1, 5) \implies |1 - 5| = 4$
- $(2, 4) \implies |2 - 4| = 2$
- Middle element 3 is ignored.

Result: $diffs = [4, 2]$. Step 1 cost: $4 + 2 = 6$.

Step 2: Symmetrize the $diffs$ array

Mirrored pairs in diffs: $(4, 2) \implies |4 - 2| = 2$.

Step 2 cost: 2.

Total Calculation: $6 + 2 = 8$.

Output: 8